WEEK 8 · RESP-100 BASIC EKG INTERPRETATION

Junctional Rhythms

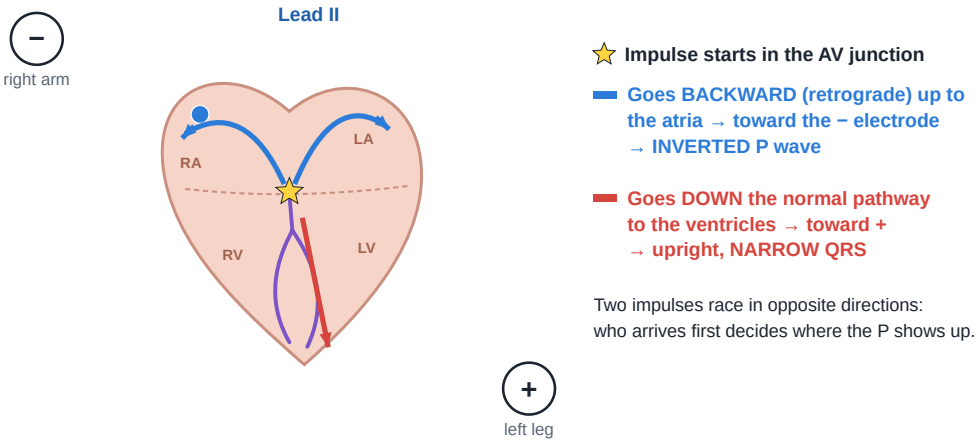
Chapter 6 · Walraven, *Basic Arrhythmias* 8th ed. · Lecture by Michael V Hyland (recording 1:08:30–1:43:14)

When the AV junction is the pacemaker, the atria fire **backward**. That gives an **inverted or missing P wave** with a narrow QRS, and the **rate** tells you which junctional rhythm it is.

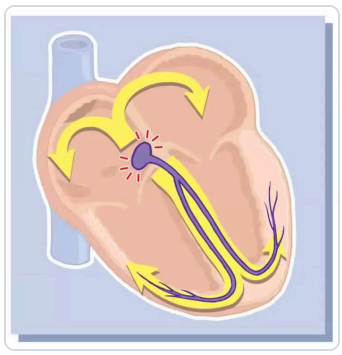
1 · Retrograde conduction | 2 · Where's the P? | 3 · PJC | 4 · Junctional escape | 5 · Accelerated junctional | 6 · Junctional tachycardia | 7 · SVT | 8 · Decision tree + cheat sheet | Quiz

1 Electrical flow from the junction

Big idea: AV node = AV junction (same thing). An impulse from the junction goes **two ways at once**: **backward (retrograde)** up to the atria and **down** the normal pathway to the ventricles. The QRS stays **narrow**, and the P wave flips **upside down**.



Lead II: the right arm is the negative electrode. Atrial depolarization heading toward it makes the P wave negative (inverted).



Slide (1:08:41): yellow arrows show flow out of the junction in both directions.

WORD TO KNOW

Retrograde = going backward. "Retrograde conduction" = electricity flowing the opposite of its normal direction (here, up into the atria).

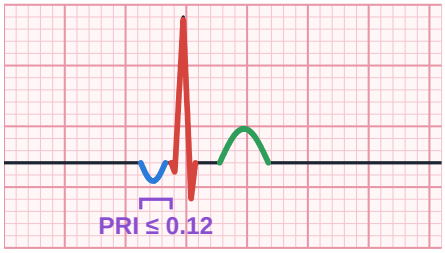
WHY THE QRS IS NARROW

The junction is still above the ventricles (supraventricular), so the impulse uses the fast His → bundle branches → Purkinje pathway.

2 Where's the P wave? It's a race

Big idea: the impulse "races" to the atria and to the ventricles. **"You beat me, I beat you, or we tie"** (1:15:17). Whoever wins decides where the inverted P shows up.

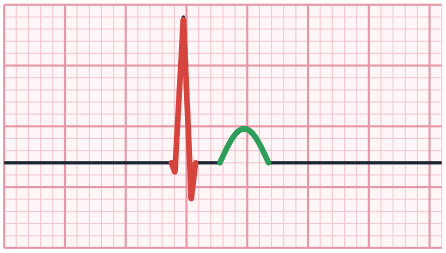
Atria first



inverted P BEFORE QRS

only time you can measure PRI

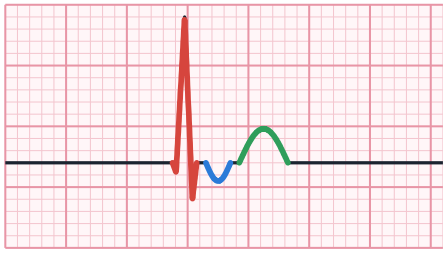
Tie (same time)



P hidden IN the QRS

PRI: can't measure

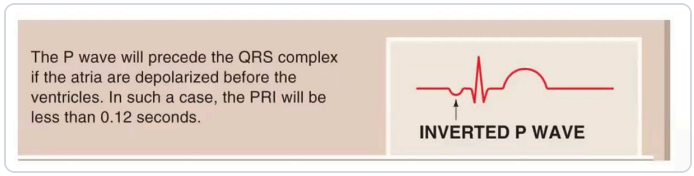
Ventricles first



inverted P AFTER QRS

PRI: can't measure

The three ways any junctional beat or rhythm can look. You can only measure a PRI when the inverted P comes **before** the QRS, and then it's short (≤ 0.12 s).



Slide (1:11:48): inverted P before the QRS, PRI < 0.12 s.

GOLDEN RULE: THE INSTRUCTOR (1:12:57)

"When you see a **narrow QRS without a P wave**, it can **only** be a junctional beat. Period. No exceptions."

And the flip side (1:17:54): an ectopic beat with a **positive (upright) P** must come from the **atria**.

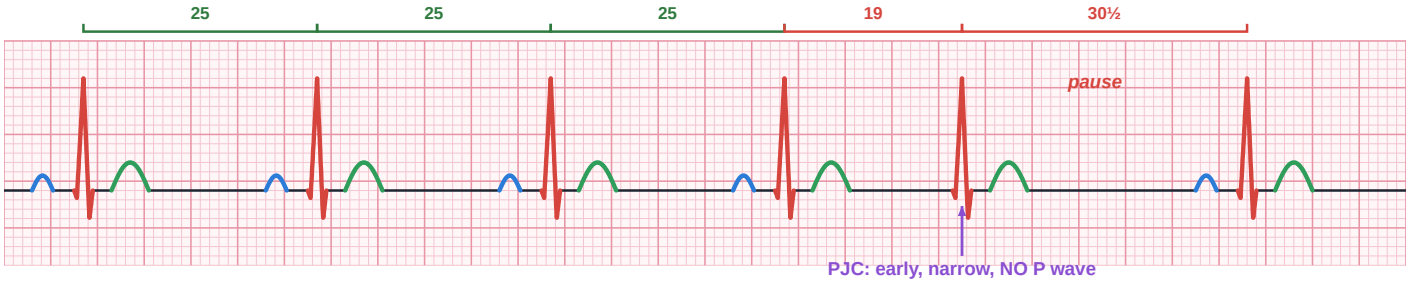
Who depolarizes first?	What you see	PRI
Atria before ventricles	P inverted, **in front of** the QRS	Measurable, **≤ 0.12 s**
Both at the same time	No visible P (hidden **inside** the QRS)	Can't measure
Ventricles before atria	P inverted, **after** the QRS (a little dip)	Can't measure

HOW TO CHART P WAVES (1:26:26)

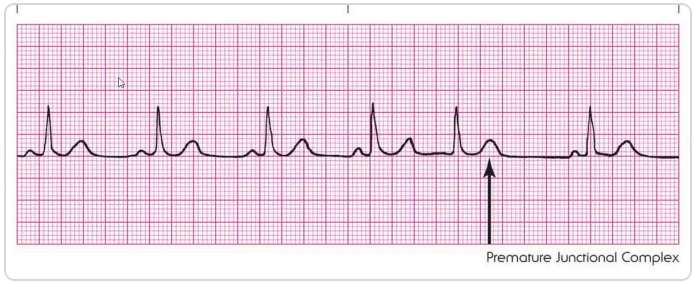
If you see them: "**inverted**". If you don't: "**none**". That's all he needs.

3 Premature junctional complex (PJC)

Big idea: the junctional version of a PAC. An irritable focus in the junction fires **once, early**. It's a single ectopic beat with a narrow QRS and an inverted or missing P. Name the underlying rhythm too, e.g. "NSR with 1 PJC."



Drawn to match the lecture strip: NSR at 60 (R-R 25 boxes). The 5th beat comes early (19 boxes) with no P wave, then there's a pause (30½).



Lecture strip (1:18:54); the arrow marks the PJC.

His worked answer

R-R 25, 25, 24, 19, 30½ → **regular w/ 1 ectopic**
Rate 1500 ÷ 25 = **60**
P: **upright & uniform** (the ectopic beat has none)
PRI 4½ boxes = **0.18** · QRS 3 boxes = **0.12**
Narrow + no P → junction → **NSR with 1 PJC**

Rules: PJC	
Regularity	Depends on the underlying rhythm
Rate	Depends on the underlying rhythm
P wave	Inverted: before, during (hidden) or after the QRS
PRI	Only if the P is in front: < 0.12 s
QRS	Narrow

PAC OR PJC?

Both are **early**, **narrow**, and followed by a pause. Look only at the early beat's **P**:

- upright but different (peaked / notched / flat) → **PAC**
- inverted or missing → **PJC**

Underlying rhythm still regular? Find **two matching R-R intervals**.

4 Junctional escape rhythm (JER)

Big idea: a **backup** rhythm. When the SA node stops or slows below the junction's own rate, the junction **escapes** and takes over at its inherent rate, **40–60 bpm**. This is a **late** rhythm, not an irritable one.

SA node
60–100 bpm · normal boss

↓

AV junction
40–60 bpm · 1st backup

↓

Ventricles
20–40 bpm · last resort

If the boss fails or slows down, the next one escapes

ESCAPE (late) beat / rhythm

backup: fires because the SA node didn't



expected beat missing → LATE junctional beat

IRRITABLE (early) beat / rhythm

fires early / faster than SA → takes over



EARLY beat (PJC), then pause

Escape: junctional escape rhythm (40–60)
Irritable: PJC, accelerated junctional, junctional tach

Pacemaker chain of command. Escape = late (the boss failed). Irritable = early or faster (someone grabbed control).



R-R 37 boxes → 1500 ÷ 37 = 40 bpm · regular · no P · narrow QRS → junctional escape rhythm

Drawn to match the lecture strip: slow, regular, narrow, no P waves.

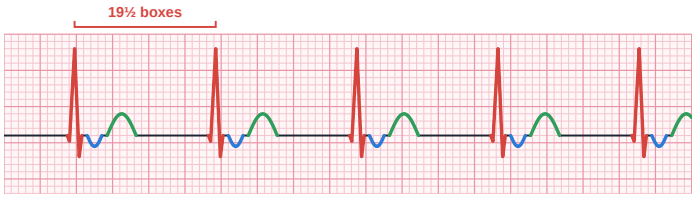


Lecture strip (1:25:10), his worked answer: R-R 37, 37, 37 → **regular** · 1500 ÷ 37 = **40** · P: **none** · PRI: can't measure · QRS 2½ boxes = **0.10** → **JER**

Rules: junctional escape	
Regularity	Regular (always)
Rate	**40–60 bpm**
P wave	Inverted before/after, or hidden in the QRS
PRI	Only if P in front: < 0.12 s
QRS	Narrow

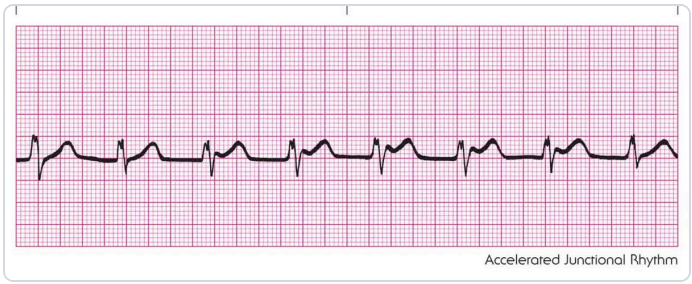
5 Accelerated junctional rhythm (AJR)

Big idea: now the junction is **irritable**. It fires **faster than its normal 40–60, at 60–100 bpm**, which beats the SA node, and "fastest rate wins." It's **not** an escape rhythm.



R-R 19½ → 77 bpm · inverted P AFTER QRS · narrow → accelerated junctional

Drawn example showing the "ventricles first" version: an inverted P **after** each QRS.



Lecture strip (1:30:31), his worked answer: R-R 19½ → **regular** · 1500 ÷ 19½ = **77** · P: **none** · QRS 3 boxes = **0.12** → **AJR**

Rules: accelerated junctional	
Regularity	Regular
Rate	**60–100 bpm**
P wave	Inverted before/after, or hidden in the QRS
PRI	Only if P in front: < 0.12 s
QRS	Narrow

SLIDE TYPO (1:29:39)

The AJR *mechanism* slide says "Rate: 100–180 bpm." **Cross it out and write 60–100.** (100–180 is junctional tach.)

"BUNNY EARS" QRS

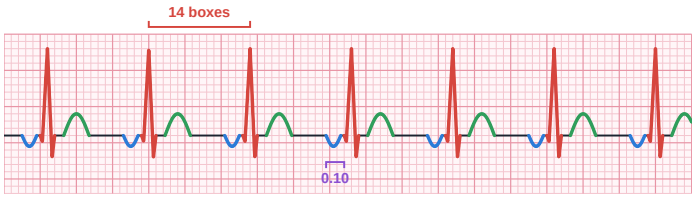
The notched R on this strip is an intraventricular conduction delay (IVCD). You don't need to name it; just measure R-R from the **same point** on each beat.

INSTRUCTOR (1:33:22)

"Just be clear on the rate. A lot of students fumble on the rate." **40–60 = junctional escape. 60–100 = accelerated junctional.**

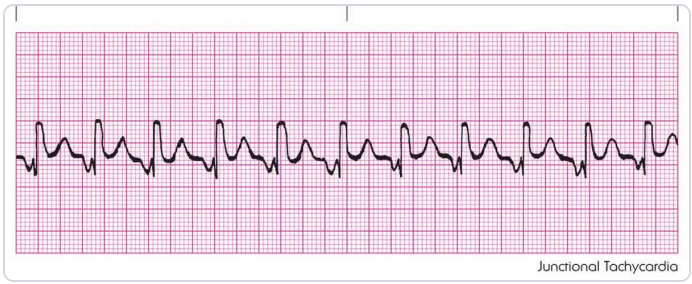
6 Junctional tachycardia

Big idea: the junction is now **very irritable** and fires **100–180 bpm**, overriding the SA node. Same P rules (inverted or missing), narrow QRS.



1500 ÷ 14 = 107 bpm · inverted P before QRS, PRI 0.10 → junctional tach

Drawn example: inverted P **in front of** each QRS, so the PRI can be measured (0.10).

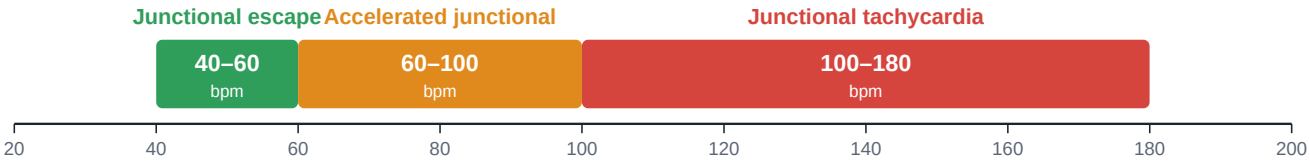


Lecture strip (1:35:50), his worked answer: R-R ≈ 14 → **regular** · 1500 ÷ 14 = **107** · P: **inverted** · PRI 2½ boxes = **0.10** · QRS 3 boxes → **junctional tach**

Rules: junctional tachycardia	
Regularity	Regular
Rate	**100–180 bpm**
P wave	Inverted before/after, or hidden in the QRS
PRI	Only if P in front: < 0.12 s
QRS	Narrow

ASIDE: ST ELEVATION (1:36:57)

On his strip the S wave never gets back to the isoelectric line before running into the T wave. That's **ST elevation**, the most common finding in an acute MI. It's a 12-lead topic, not part of this interpretation.

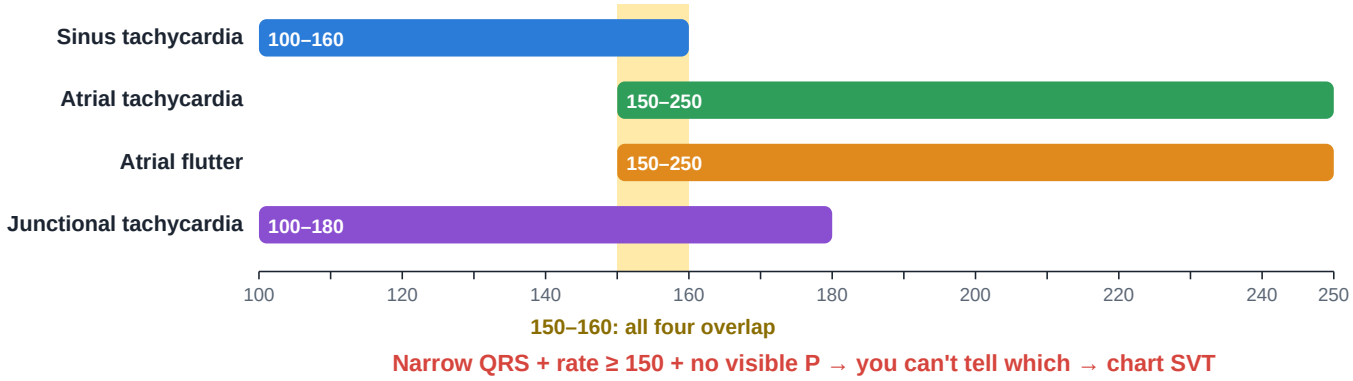


Same strip look (regular, narrow, inverted/absent P): the RATE picks the name. All junctional rhythms are ALWAYS regular.

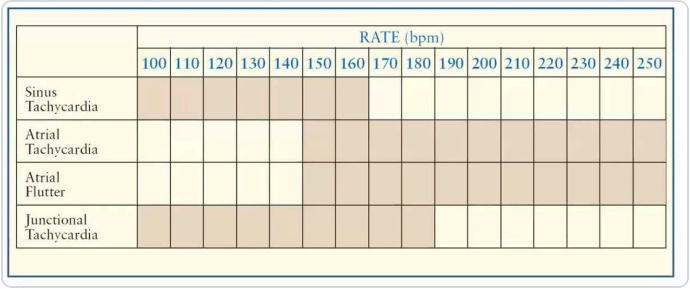
The three junctional rate ranges you must know (1:35:06). "Junctional rhythms are always regular. Yes, always."

7 Supraventricular tachycardia (SVT)

Big idea: SVT is a **catch-all name**, not a separate pacemaker. Use it for a **rapid, regular, narrow-complex** rhythm when you **can't see P waves** and the rate fits several rhythms, so you can't tell where it's coming from.



Overlapping rate ranges (from the slide at 1:41:11). Without P waves, a narrow rhythm at 155 could be any of the four.



	RATE (bpm)														
	100	110	120	130	140	150	160	170	180	190	200	210	220	230	240	250
Sinus Tachycardia	■	■	■	■	■	■	■									
Atrial Tachycardia						■	■	■	■	■	■	■	■	■	■	■
Atrial Flutter						■	■	■	■	■	■	■	■	■	■	■
Junctional Tachycardia	■	■	■	■	■	■	■	■	■							

Textbook table: shaded cells = possible rate.

WHAT HE EXPECTS (1:42:59)

"When you get **narrow complex** with heart rates **above 150** that you **cannot see the P wave**, your interpretation should be **SVT**."

- 150–180: sinus tach, atrial tach, flutter and junctional tach can all overlap
- Above 180: only atrial tach or flutter (up to 250)

Narrow · 107 · inverted P

You can see where it comes from → **junctional tach**, not SVT.

Narrow · 214 · no P

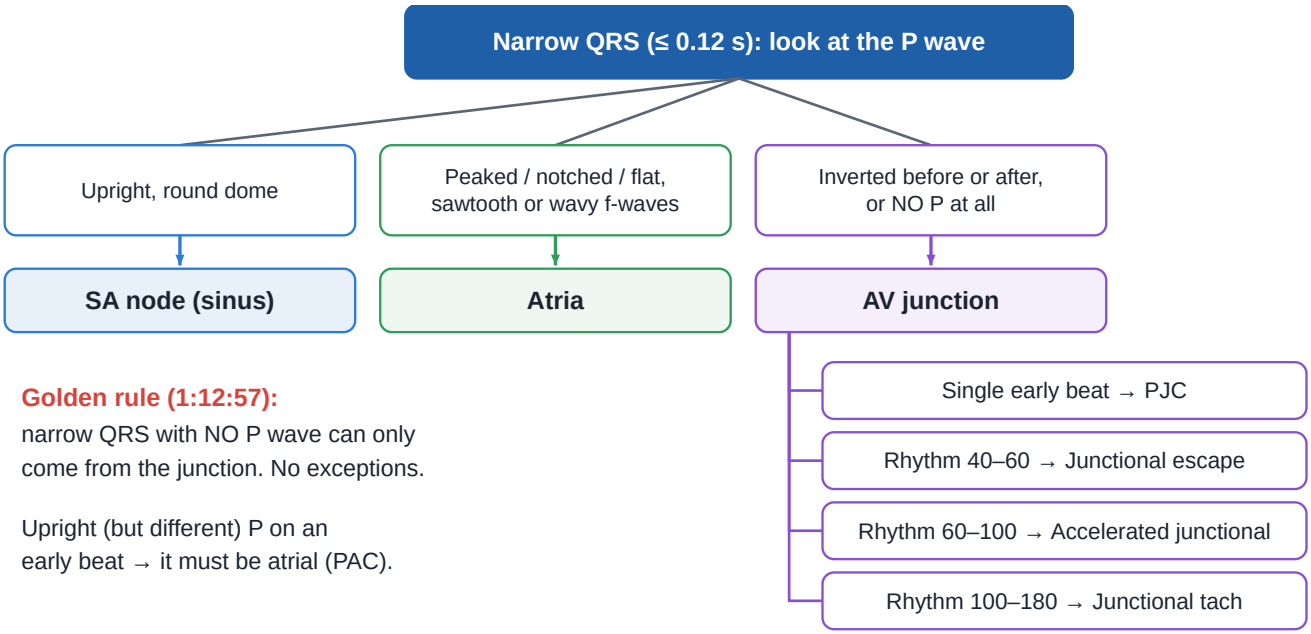
Can't see the P → **SVT** (atrial tach accepted).

Narrow · 155 · no P

Four possible rhythms → **SVT**.

8 Putting it together

Big idea: narrow QRS? Let the **P wave** tell you the origin. If it's junctional, let the **rate** (or "single early beat") tell you the name.



Golden rule (1:12:57):
narrow QRS with NO P wave can only come from the junction. No exceptions.

Upright (but different) P on an early beat → it must be atrial (PAC).

From P wave to origin to rhythm name.

Cheat sheet: Chapter 6 junctional rhythms

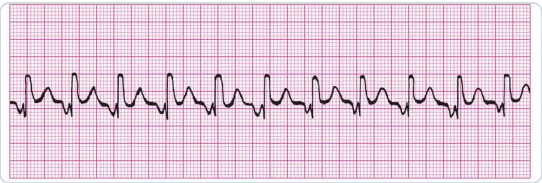
Rhythm	Regularity	Rate	P wave	PRI	QRS	Key clue
PJC	Underlying + 1 ectopic	Underlying	Ectopic: inverted or none	< 0.12 if P in front	Narrow	Early beat, no/inverted P, pause
Junctional escape	Regular	**40–60**	Inverted or none	< 0.12 if P in front	Narrow	Backup (late) rhythm
Accelerated junctional	Regular	**60–100**	Inverted or none	< 0.12 if P in front	Narrow	Irritable; slide typo says 100–180
Junctional tachycardia	Regular	**100–180**	Inverted or none	< 0.12 if P in front	Narrow	Fast + inverted P
SVT (catch-all)	Regular	≥ 150	Not visible	Can't measure	Narrow	Can't tell the origin

MEMORY TRICK: JUNCTIONAL RATES GO "4-6-1-8"

40 … **60** … **100** … **180**: escape | accelerated | tachy. Same picture, three speeds. **Next test (0:16:44):** at least one strip each of PJC, JER, AJR and junctional tach.

? Self-check quiz

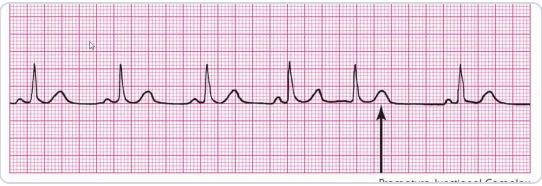
1. Why are junctional P waves inverted in lead II? Use the words *retrograde* and *negative electrode*.
2. Name the three places a junctional P wave can appear, and say when each happens.
3. When can you measure the PRI in a junctional rhythm, and what will it be?
4. Complete the instructor's rule: "A narrow QRS with no P wave can only come from …"
5. An early, narrow beat has an upright, notched P. PAC or PJC? What if it had no P?
6. What's the difference between an *escape* rhythm and an *irritable* rhythm? Which junctional rhythm is the escape one?
7. Give the rate ranges for JER, AJR and junctional tachycardia. Which slide has a typo?
8. Regular, narrow QRS, R-R 30 small boxes, no P waves. Rate? Interpretation?
9. Regular, narrow QRS, R-R 20 small boxes, inverted P after every QRS. Rate? Interpretation?
10. Narrow QRS at 165 bpm, regular, no visible P. Interpretation, and why?
11. Identify strips A–D below.



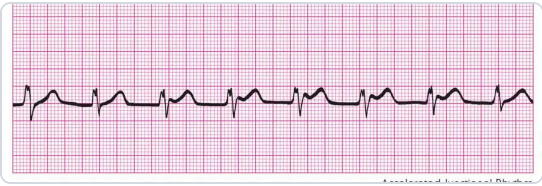
A



B



C



D

Answers. 1. The impulse travels retrograde (backward) from the junction up to the atria, toward the right-arm negative electrode, so the P is negative. 2. Before the QRS (atria first), hidden in the QRS (same time), after the QRS (ventricles first). 3. Only when the inverted P is in front of the QRS; it will be short, ≤ 0.12 s. 4. "…the junction. Period. No exceptions." 5. Upright notched P → PAC. No P → PJC. 6. Escape = backup that fires late because the SA node failed or slowed. Irritable = fires early or faster and takes over. JER is the escape rhythm. 7. JER 40–60, AJR 60–100, JT 100–180. The AJR mechanism slide wrongly says 100–180. 8. 1500 ÷ 30 = 50 → junctional escape rhythm. 9. 1500 ÷ 20 = 75 → accelerated junctional rhythm. 10. SVT: narrow-complex tachycardia ≥ 150 with no visible P, and sinus tach, atrial tach, flutter and junctional tach all overlap there. 11. A = junctional tachycardia (107, inverted P). B = junctional escape rhythm (40). C = NSR with 1 PJC. D = accelerated junctional rhythm (77).